https://github.com/TSika-1/CS352-Unit3-Assignment2.git

State of the project. Everything works. The no-parameter constructor builds the hardcoded cave, toString prints it, and solve finds the mirror pool using a depth-first search in four directions. getPath returns the directions (wwsse for the hardcoded cave) or an empty string if there is no path. The file constructor reads the cave from testdat.txt. I have no known bugs.

How I tested. I first ran the hardcoded cave and checked that the path wwsse matched the example in the assignment. Then I tested a cave from a file with a longer path that turns in more than one direction:

6 7
RRRRRRR
RS...RR
RRRR.RR
RRRR.RR
RM...RR
RRRRRRR

This cave needs 3 moves east, 3 south, and 3 west, and the program returned eeesssswww. An early test with a tiny 2×3 cave only needed one move, so it didn't really prove the search worked. That pushed me to build the longer layout. [Add anything that actually made you change your code, such as a bug you hit.]

What I learned. I learned how to use recursion with backtracking to search a grid. I also learned to mark visited cells so the search never loops, and to undo the mark at dead ends.

What I liked. I liked seeing the final layout with the path drawn on it, which made it easy to check that the answer was right.

What was confusing. It took me a while to decide how to avoid infinite loops and where to undo my marks when backing out of a dead end. It would help if the assignment gave an example of a cave with no path to test that case.

What I'd add with more time. I would add a cave with no path to test that case, and I would let the program handle caves with more than one path by finding the shortest one.